

Classification Tree

Lecture 2

Nirjhor Datta

What is Classification?

Classification means predicting a **category** or **class label**.

Examples

- Email: Spam or Not Spam
- Student: Pass or Fail
- Patient: Disease or No Disease
- Customer: Will Buy or Will Not Buy

Key Point

The output of classification is a class, not a continuous number.

Motivating Example

Suppose we want to predict whether a person will like the movie **Troll 2**.

Loves Popcorn	Loves Soda	Age	Loves Troll 2?
Yes	Yes	7	No
Yes	No	12	No
No	Yes	18	Yes
Yes	Yes	35	Yes
No	No	38	No
No	Yes	50	Yes
Yes	No	83	No

Problem

Given a new person, can we predict whether they will love Troll 2?

A Human Way of Thinking

A human may ask simple questions:

- Does the person love soda?
- How old is the person?
- Does the person love popcorn?

Then we make a decision based on the answers.

Example

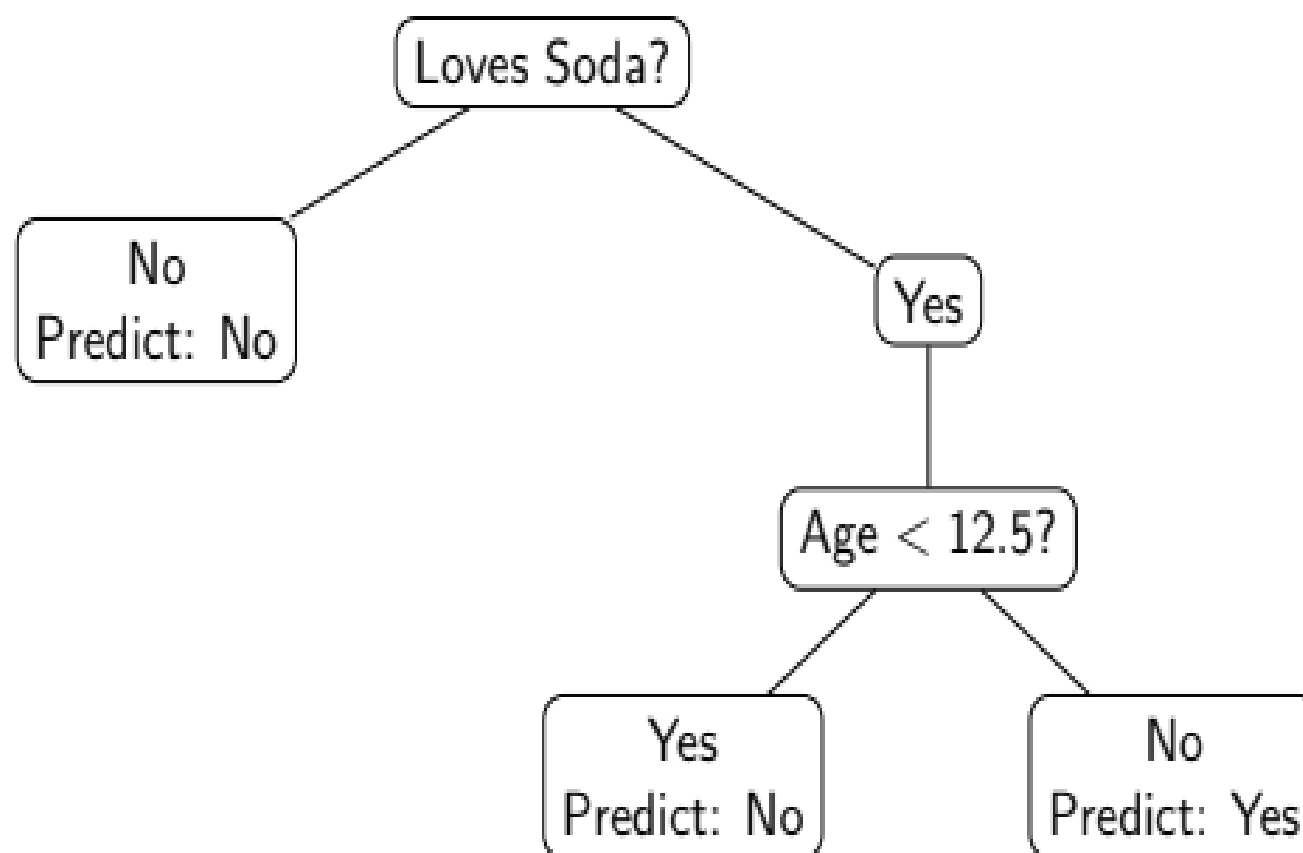
If someone loves soda and is older than 12.5, maybe they are likely to love Troll 2.

This is the basic idea of a classification tree.

A classification tree predicts by asking a sequence of questions.

What is a Classification Tree?

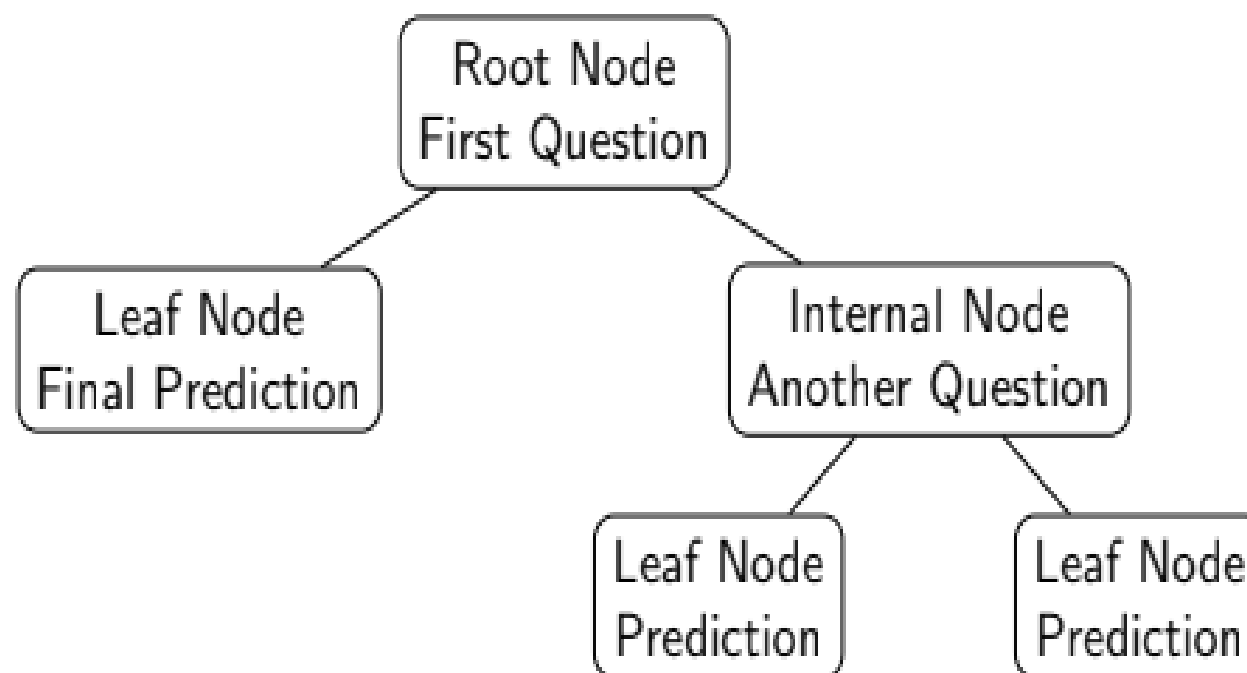
A classification tree is a tree-shaped model used for classification.



How to Read It

Start from the top, answer the questions, and stop at a prediction.

Parts of a Classification Tree



- **Root node:** first question
- **Internal node:** intermediate question
- **Leaf node:** final class prediction

Prediction Using a Tree

Suppose a new person has:

Feature	Value
Loves Soda	Yes
Age	18
Loves Popcorn	No

Using the tree:

- 1 Loves Soda? Yes
- 2 Age < 12.5? No
- 3 Predict: Yes

Final Prediction

The person is predicted to love Troll 2.

The Important Question

Now we understand how to use a tree.

But how do we **build** the tree?

Which question should be asked first?

Possible first questions:

- Loves Popcorn?
- Loves Soda?
- Age < 15?
- Age < 36.5?

Need

We need a rule to choose the best question.

What Makes a Good Question?

A good question separates the data into cleaner groups.

Before Split

3 Yes, 4 No

This is mixed.

After a Good Split

Group 1: Mostly No

Group 2: Mostly Yes

This is easier to classify.

Main Idea

The best split makes the child nodes more pure.

Purity and Impurity

A node is **pure** if all or most samples belong to one class.

Node	Class Distribution	Quality
A	3 Yes, 4 No	Impure
B	3 Yes, 0 No	Pure
C	0 Yes, 4 No	Pure
D	2 Yes, 2 No	Very Impure

Goal of Tree Construction

Choose splits that reduce impurity.

How Can We Measure Impurity?

There are two common ways:

Measure	Meaning
Gini Impurity	Measures how mixed a node is
Entropy	Measures uncertainty in a node
Information Gain	Measures how much uncertainty decreases after a split

Important

These formulas help the tree decide which split is best.

Gini Impurity

For a two-class problem:

$$Gini = 1 - p_{Yes}^2 - p_{No}^2$$

where:

$$p_{Yes} = \frac{\text{Number of Yes samples}}{\text{Total samples}}$$

$$p_{No} = \frac{\text{Number of No samples}}{\text{Total samples}}$$

Interpretation

- Gini = 0 means perfectly pure.
- Higher Gini means more mixed.

Example: Gini Before Splitting

Before splitting, the dataset has:

3 Yes, 4 No

So:

$$p_{Yes} = \frac{3}{7}, \quad p_{No} = \frac{4}{7}$$

$$Gini = 1 - \left(\frac{3}{7}\right)^2 - \left(\frac{4}{7}\right)^2$$

$$Gini = 1 - \frac{9}{49} - \frac{16}{49}$$

$$Gini = \frac{24}{49} \approx 0.49$$

Meaning

The original dataset is mixed.

Weighted Gini After Split

After a split, we calculate the impurity of each child node.

$$Gini_{after} = \frac{n_1}{n} Gini_1 + \frac{n_2}{n} Gini_2$$

- n = total samples
- n_1 = samples in child 1
- n_2 = samples in child 2

Rule

The best split has the lowest weighted Gini after splitting.

Entropy

Entropy measures uncertainty.

$$Entropy = -p_{Yes} \log_2(p_{Yes}) - p_{No} \log_2(p_{No})$$

Interpretation

- Entropy = 0 means pure.
- Entropy is high when classes are mixed.

Before Split

For 3 Yes and 4 No:

$$Entropy \approx 0.985$$

Information Gain

Information Gain tells us how much uncertainty is reduced after a split.

$$\text{Information Gain} = \text{Entropy}_{\text{before}} - \text{Entropy}_{\text{after}}$$

Rule

Choose the split with the highest information gain.

Higher Information Gain = Better Split

Choosing the Root Node

For the full dataset:

$$Entropy_{before} = 0.985$$

Now calculate information gain for possible first splits.

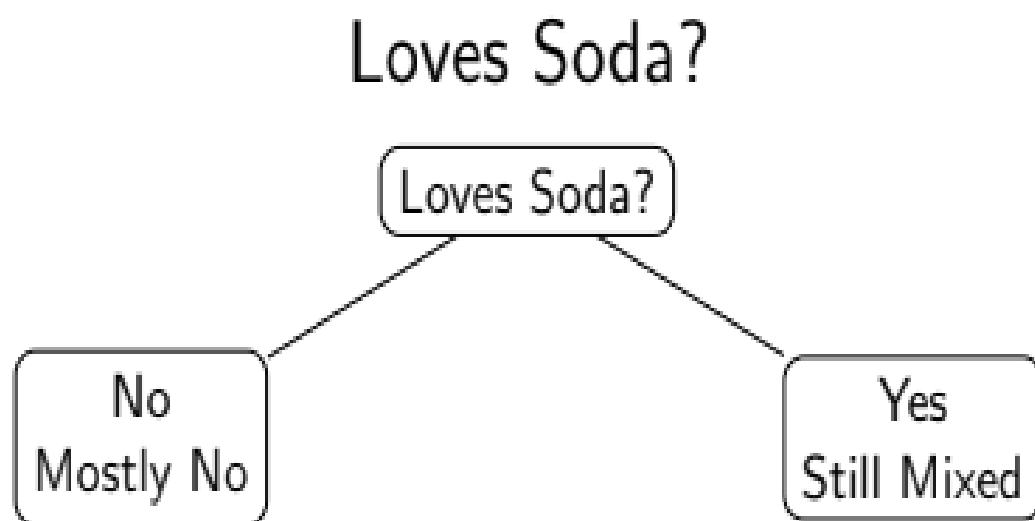
Possible Split	Information Gain
Loves Popcorn	0.128
Loves Soda	0.521
Age < 15	0.291

Decision

Since **Loves Soda** has the highest information gain, it becomes the root node.

First Split

The first question becomes:



Next Step

For any branch that is still mixed, repeat the same process.

Choosing the Next Split

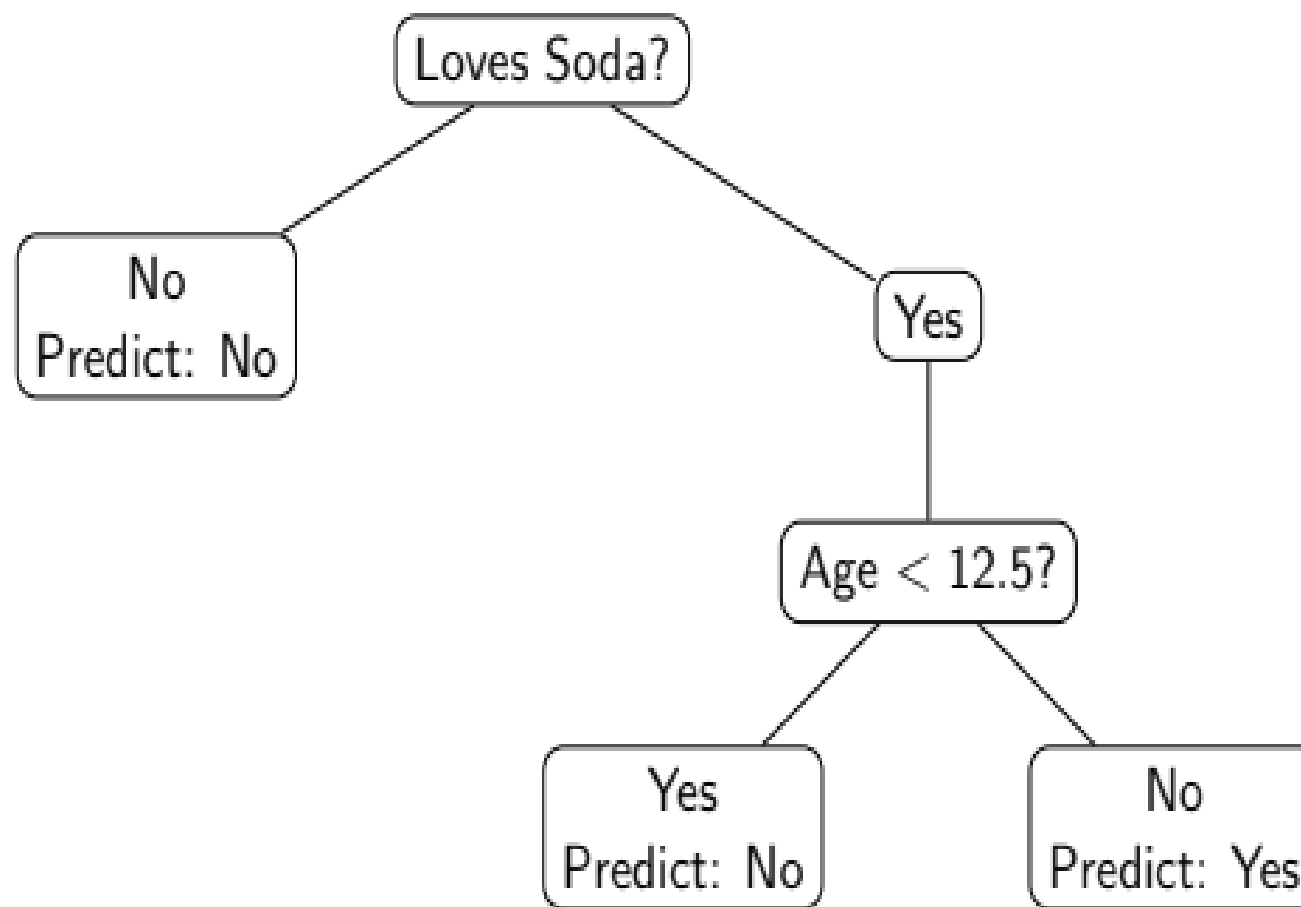
Inside the mixed branch, we again test possible questions.

Possible Split	Information Gain
Loves Popcorn	0.311
Age < 12.5	0.811

Decision

Since **Age** < 12.5 has the highest information gain, it becomes the next question.

Final Classification Tree



Final Model

This tree can now classify new examples.

Tree as Decision Rules

The tree can also be written as rules.

Rule 1

If Loves Soda = No, then Predict No.

Rule 2

If Loves Soda = Yes and Age < 12.5 , then Predict No.

Rule 3

If Loves Soda = Yes and Age ≥ 12.5 , then Predict Yes.

Another Prediction Example

New person:

Feature	Value
Loves Soda	Yes
Age	35
Loves Popcorn	Yes

Using the tree:

- 1 Loves Soda? Yes
- 2 Age < 12.5? No
- 3 Predict: Yes

Prediction

The person is predicted to love Troll 2.

Why Classification Trees Are Useful

- Easy to understand
- Easy to explain visually
- Can handle categorical and numerical features
- Produces human-readable rules
- Useful for both teaching and practical classification problems

Main Strength

Classification trees are highly interpretable.

Limitations

Classification trees are simple and interpretable, but they have limitations.

- They can overfit the training data.
- Very deep trees may become complex.
- Small changes in data can change the tree.
- A single tree may not always give the best accuracy.

Later Idea

Methods like Random Forest use many trees together to reduce these problems.

Class Activity

Ask students to build the first split.

Weather	Humidity	Windy	Play?
Sunny	High	No	No
Sunny	High	Yes	No
Overcast	High	No	Yes
Rain	High	No	Yes
Rain	Normal	No	Yes
Rain	Normal	Yes	No
Overcast	Normal	Yes	Yes
Sunny	High	No	No

Question

Which feature seems like the best root node: Weather, Humidity, or Windy?

ID3: Basic Idea

ID3 stands for **I**terative **D**ichotomiser **3**.

Purpose

ID3 is an algorithm used to construct a classification tree.

- It builds the tree from the top.
- At each node, it chooses the best attribute for splitting.
- The best attribute is selected using **Information Gain**.
- The process continues until the data becomes pure or no useful split remains.

Main Intuition

ID3 asks:

Which question reduces uncertainty the most?

How ID3 Chooses the Best Split

ID3 uses **Entropy** and **Information Gain**.

$$Entropy(S) = -p_{Yes} \log_2(p_{Yes}) - p_{No} \log_2(p_{No})$$

Entropy measures how mixed or uncertain a node is.

$$Gain(S, A) = Entropy(S) - Entropy(S|A)$$

Information Gain measures how much uncertainty decreases after splitting by attribute A .

ID3 Rule

Choose the attribute with the highest Information Gain.

$$\text{Best Attribute} = \arg \max_A Gain(S, A)$$

Class Activity Solution: ID3 Algorithm

ID3 builds a decision tree by choosing the attribute with the highest **Information Gain**.

$$Gain(S, A) = Entropy(S) - Entropy(S|A)$$

where:

$$Entropy(S|A) = \sum_{v \in Values(A)} \frac{|S_v|}{|S|} Entropy(S_v)$$

Dataset Summary

Total examples = 8

Yes = 4, No = 4

$$Entropy(S) = -\frac{4}{8} \log_2 \frac{4}{8} - \frac{4}{8} \log_2 \frac{4}{8} = 1$$

Information Gain for Weather

Weather has three possible values:

$$\text{Weather} \in \{\text{Sunny}, \text{Overcast}, \text{Rain}\}$$

Weather	Yes	No	Entropy
Sunny	0	3	0
Overcast	2	0	0
Rain	2	1	0.918

$$\text{Entropy}(S|\text{Weather}) = \frac{3}{8}(0) + \frac{2}{8}(0) + \frac{3}{8}(0.918)$$

$$\text{Entropy}(S|\text{Weather}) = 0.344$$

$$\text{Gain}(S, \text{Weather}) = 1 - 0.344 = 0.656$$

Information Gain for Humidity

Humidity has two possible values:

$$\text{Humidity} \in \{\text{High}, \text{Normal}\}$$

Humidity	Yes	No	Entropy
High	2	3	0.971
Normal	2	1	0.918

$$\text{Entropy}(S|\text{Humidity}) = \frac{5}{8}(0.971) + \frac{3}{8}(0.918)$$

$$\text{Entropy}(S|\text{Humidity}) = 0.951$$

$$\text{Gain}(S, \text{Humidity}) = 1 - 0.951 = 0.049$$

Information Gain for Windy

Windy has two possible values:

$$Windy \in \{No, Yes\}$$

Windy	Yes	No	Entropy
No	3	2	0.971
Yes	1	2	0.918

$$Entropy(S|Windy) = \frac{5}{8}(0.971) + \frac{3}{8}(0.918)$$

$$Entropy(S|Windy) = 0.951$$

$$Gain(S, Windy) = 1 - 0.951 = 0.049$$

Choosing the Root Node

Now compare all information gains:

Attribute	Information Gain
Weather	0.656
Humidity	0.049
Windy	0.049

ID3 Decision

Since **Weather** has the highest information gain, ID3 selects **Weather** as the root node.

Root = Weather

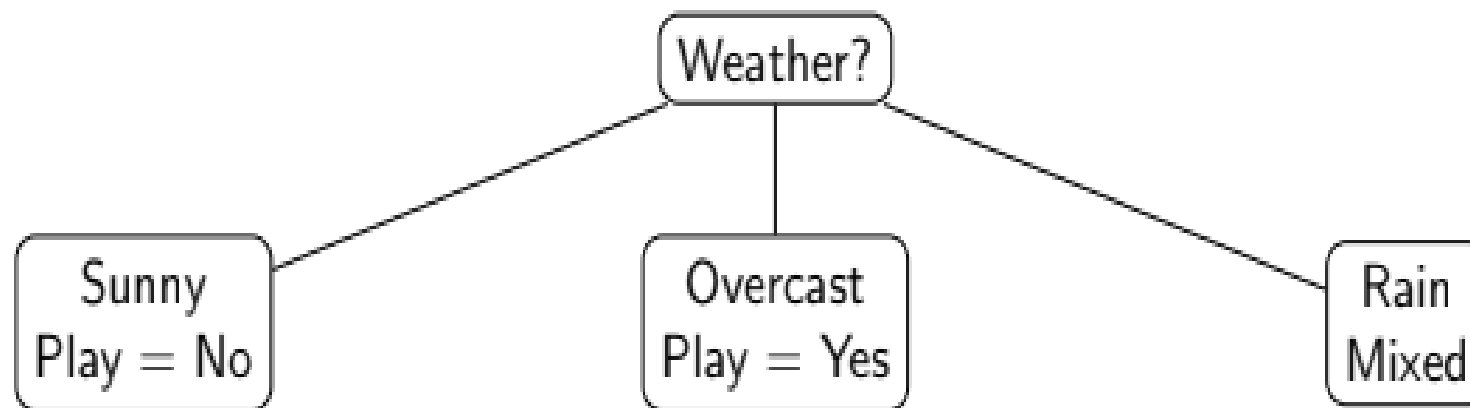
Tree After First Split

After choosing **Weather** as the root:

Weather = Sunny $\Rightarrow$ Play = No

Weather = Overcast $\Rightarrow$ Play = Yes

Weather = Rain $\Rightarrow$ Still mixed



Next Step

Only the Rain branch needs another split.

Solving the Rain Branch

For the Rain subset:

Humidity	Windy	Play?
High	No	Yes
Normal	No	Yes
Normal	Yes	No

For this subset:

$$Yes = 2, \quad No = 1$$

$$Entropy(Rain) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.918$$

Now ID3 tests the remaining attributes:

Humidity, Windy

Rain Branch: Gain for Humidity

Inside the Rain branch:

Humidity	Yes	No	Entropy
High	1	0	0
Normal	1	1	1

$$Entropy(Rain|Humidity) = \frac{1}{3}(0) + \frac{2}{3}(1)$$

$$Entropy(Rain|Humidity) = 0.667$$

$$Gain(Rain, Humidity) = 0.918 - 0.667 = 0.251$$

Rain Branch: Gain for Windy

Inside the Rain branch:

Windy	Yes	No	Entropy
No	2	0	0
Yes	0	1	0

$$\text{Entropy}(\text{Rain}|\text{Windy}) = \frac{2}{3}(0) + \frac{1}{3}(0) = 0$$

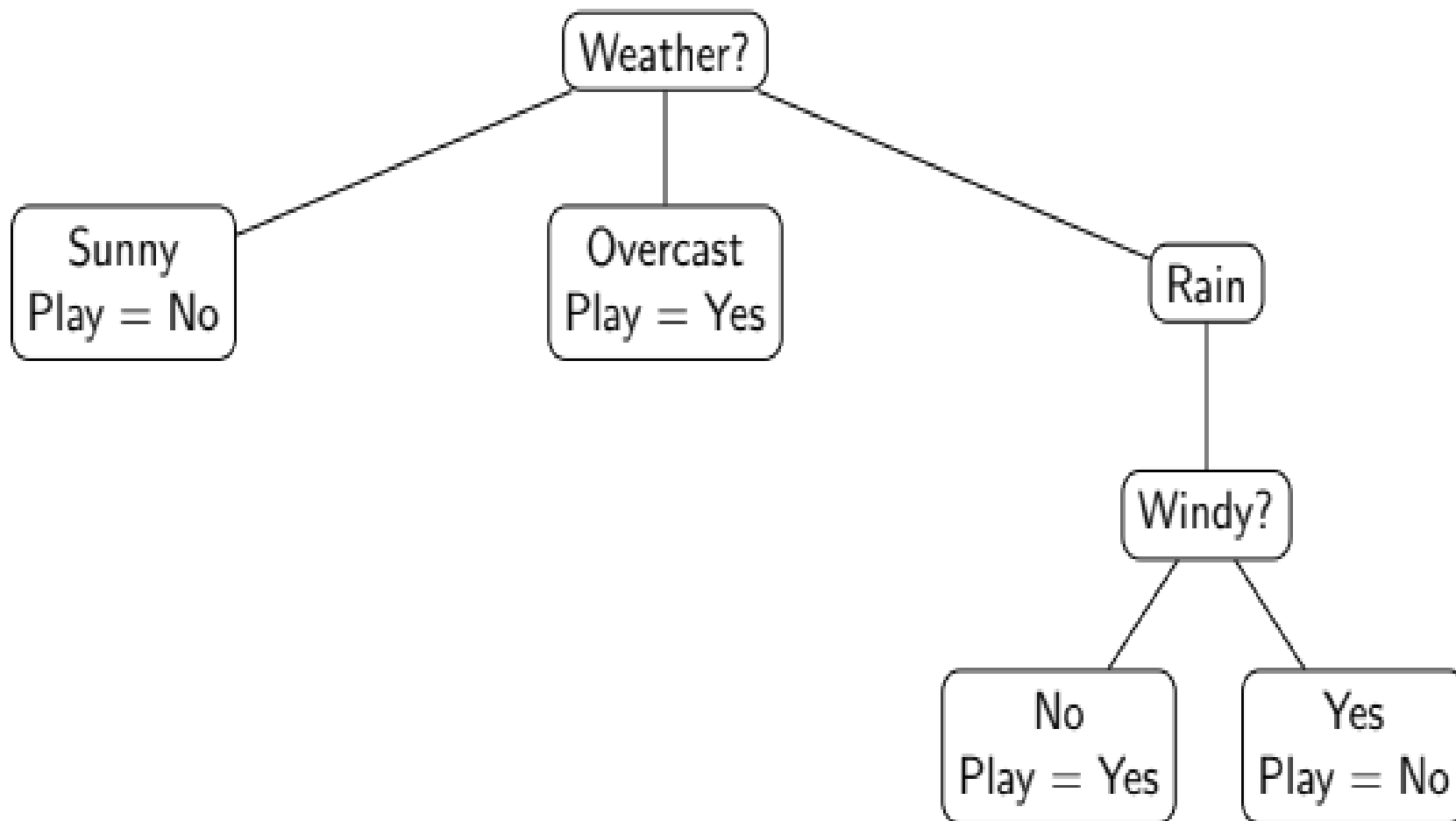
$$\text{Gain}(\text{Rain}, \text{Windy}) = 0.918 - 0 = 0.918$$

ID3 Decision

Inside the Rain branch, ID3 chooses **Windy**.

Rain branch split = Windy

Final ID3 Classification Tree



Final Tree

The root node is **Weather**. The Rain branch is further split using **Windy**.

Final Rules from the ID3 Tree

The final decision tree can be written as simple rules:

Rule 1

If Weather = Sunny, then Play = No.

Rule 2

If Weather = Overcast, then Play = Yes.

Rule 3

If Weather = Rain and Windy = No, then Play = Yes.

Rule 4

If Weather = Rain and Windy = Yes, then Play = No.

ID3 Solution Summary

- ID3 uses entropy and information gain.
- The attribute with the highest information gain becomes the split.
- For the full dataset, **Weather** has the highest gain.
- Therefore, **Weather** becomes the root node.
- The Rain branch is still mixed.
- Inside the Rain branch, **Windy** gives the highest gain.

Weather → *Windy*

This is the final classification tree using ID3.

Summary

- Classification predicts class labels.
- A classification tree predicts by asking questions.
- The first question is called the root node.
- Leaf nodes give final predictions.
- A good split makes child nodes more pure.
- Gini, entropy, and information gain help choose good splits.

Classification Tree = Questions + Splits + Final Class

Questions?